PRANAY LACHULURI

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EDUCATION

University at Buffalo, SUNY, MS in Computer Science (AI/ML) | GPA 3.7/4.0 — Aug 2025 – Jan 2027 (expected)
Coursework: Deep Learning, Intro to Machine Learning, Computer Vision & Image Processing, Statistical Data Mining II, Data-Intensive Computing, Computer Security, Operating Systems, Analysis of Algorithms, Data Models & Query Languages

Neil Gogte Institute of Technology, BE in Computer Science (AI/ML) | CGPA 7.1/10 — 2020 – 2024 | Hyderabad, India

EXPERIENCE

Graduate Research Assistant, University at Buffalo, ASAP (Autonomous Plastic-Sort) — Sep 2026 – Present | Buffalo, NY

- Building the data foundation for a low-cost optical/infrared plastic-identification system, operating a multi-sensor rig of hyperspectral, RGB and thermal cameras and a 4-chip mmWave radar on ROS 2.
- Perform hyperspectral calibration (dark/white references, flat-field reflectance correction), multi-pose checkerboard calibration across RGB, thermal and hyperspectral views, and mmWave range-bias calibration.
- Capture synchronized static and conveyor-belt recordings of PET, HDPE and LDPE across 10 controlled light levels and several belt speeds; set up CVAT and FiftyOne for annotation and dataset management.

Apprentice, Google/YouTube Client Operations, Genpact — Mar 2025 – Aug 2025 | India

- Performed data validation and quality-focused operations for a Google/YouTube client under strict client-defined policies and quality standards.
- Trained in generative AI, agentic AI and AI-assisted, prompt-driven workflows.

Data Science Intern, YHills — Apr 2023 – Jul 2023 | Remote

- Built and compared 7 scikit-learn and XGBoost models per task: fraud detection at **99.2%** accuracy (XGBoost), H1N1 vaccination prediction at **82.6%**, and taxi-fare regression at **R^2 0.76** (Random Forest).

PROJECTS

ShadowFix AI: Shadow Removal | PyTorch, U-Net, OpenCV, Gradio — 2026

- Built a 7.76M-parameter U-Net and compared RGB, residual and mask-guided variants on ISTD (1,197 train / 133 val / 540 test images) with AMP training and PSNR/SSIM plus shadow-region metrics.
- Best mask-guided model reached **30.19 dB PSNR** and **0.951 SSIM**, **+9.63 dB** over the raw input; shipped automatic and paint-the-shadow Gradio modes.

Trash Detection with YOLOv8 | YOLOv8, PyTorch, OpenCV, Gradio — 2026

- Fine-tuned YOLOv8m on a 59-class TACO-derived dataset at 832 px with AdamW, cosine LR, Mosaic and MixUp; 0.183 mAP@50 on 299 validation images (962 objects), +44% mAP@50-95 over a YOLOv8n baseline.
- Added confidence-weighted trash heatmaps, a 0–100 cleanliness score and an interactive Gradio assistant.

Transformers from Scratch: ViT and GPT | PyTorch — 2026

- Implemented a Vision Transformer and a character-level GPT from scratch; ablations on CIFAR-10 showed 2×2 patches beat 8×8 by 8.9 points (49.3% vs 40.4%) and positional encoding adds 3.2 points.

AI Security: Adversarial and LLM Attacks | PyTorch, Hugging Face Transformers — 2025

- Ran FGSM and PGD attacks on a ResNet-50 classifier, jailbreak attacks on StableLM-3B and indirect prompt injection on Qwen2.5, and evaluated defences.

Real-time Sign Language Detection | TensorFlow/Keras, MediaPipe, OpenCV — 2024

- Built a webcam pipeline that extracts MediaPipe hand landmarks and classifies gestures with a custom Keras model, with stable live predictions on screen.

Car Price Prediction App | scikit-learn, Flask — 2023

- Served a one-hot-encoded linear regression pipeline through a Flask app with dynamic inputs and input validation; reproducible training script with cross-validated R^2 of about 0.64.

SKILLS

Languages: Python, SQL, Java, JavaScript

ML / Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, CNNs, U-Net, Vision Transformers, YOLO, transfer learning, Hugging Face Transformers

Computer Vision & Sensing: OpenCV, MediaPipe, object detection, image restoration, hyperspectral and thermal imaging, mmWave radar, camera calibration

Data & Tools: PySpark, pandas, NumPy, MySQL, Git, Linux, Docker, ROS 2, Flask, Gradio, CVAT, FiftyOne, Jupyter, Colab